

Intermediate Elective 2

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Set-up

Read-in packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(Cairo)

# customization
library(prismatic)
library(extrafont)
library(scales)
library(stringr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

## 1. LOAD PACKAGES & SETUP ----
suppressPackageStartupMessages({
  if (!require("pacman")) install.packages("pacman")
})
```

```

pacman::p_load(
  tidyverse, # Easily Install and Load the 'Tidyverse'
  ggtext,    # Improved Text Rendering Support for 'ggplot2'
  showtext,  # Using Fonts More Easily in R Graphs
  janitor,   # Simple Tools for Examining and Cleaning Dirty Data
  scales,    # Scale Functions for Visualization
  glue,      # Interpreted String Literals
  gghighlight # Highlight Lines and Points in 'ggplot2'
)})

### /- figure size ----
camcorder::gg_record(
  dir      = here::here("temp_plots"),
  device   = "png",
  width    = 8,
  height   = 8,
  units    = "in",
  dpi      = 320
)

```

Read-in data

```

# WEEK 4 data

veg <- read_csv(here("data", "veg.csv"))

metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# WEEK 5 data
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

```

```
# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
                        sheet = "Water Quality",
                        na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# WEEK 4

# create a new object from veg
pools_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP") &
    site == "ncos") |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)
  ) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats
  mutate(mean_pc = sum_pc/num_quad)

# WEEK 5

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
  )

```

```

    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
       site_full = fct_rev(site_full))

```

1. Explain your inspiration

In 1-2 sentences each, describe:

The visualization I chose for my inspiration was [Steven Ponce's "Evolution of MTA Art](#)

[Materials \(1980-2020\)](#)". This makes sense for the data sets that I am utilizing because I am trying to find the relationship between vegetation and its explanatory variables as well as invertebrate abundance and its explanatory variables. The circular graph will be able to show the abundance of both vegetation and invertebrates, as well as a statistical summary of the mean.

- which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.)

2. Plan your figure

Sketch what your figure would look like following your inspiration.

Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

Clean/wrangle your data as needed.

Then, code the figure you decided to create.

Make sure all your code is annotated, and that the output is showing.

At the end of this section, include code to save your final figure as a .jpg or .png file.

Vegetation

Vegetation graphs code

```
# create new object from veg
native_spp <- veg |>
# filter only native observations
  filter(cover_category == "NATIVE COVER") |>
# pull to extract data frame column as a vector
  pull(psoc)

# random sample
sample(native_spp, 10)
```

```
[1] "Eleocharis_acicularis"      "Spergularia_marina"
[3] "Juncus_bufonius"           "Salicornia_pacifica"
[5] "Epilobium_brachycarpum"    "Symphyotrichum_subulatum"
[7] "Eryngium_vaseyi"           "Festuca_microstachys"
[9] "Epilobium_brachycarpum"    "Heterotheca_grandiflora"
```

```
# create new object from pools_explicit0
species_rich <- pools_explicit0 |>
# group by year and pool
group_by(year, transect_name) |>
# filter data frame with vector \nto get native species,
# also filter percent cover to identify species presence
filter(psoc %in% native_spp, mean_pc > 0) |>
# species richness calculation
summarize(spp_rich = n_distinct(psoc)) |>
mutate(transect_name = case_when(
  transect_name == "VP-01" ~ "Pool 1",
  transect_name == "VP-02" ~ "Pool 2",
  transect_name == "VP-03" ~ "Pool 3",
  transect_name == "VP-04" ~ "Pool 4",
  transect_name == "VP-05" ~ "Pool 5",
  transect_name == "VP-06" ~ "Pool 6",
  transect_name == "VP-07" ~ "Pool 7",
  transect_name == "VP-08" ~ "Pool 8"
))
# # ungroup data frame
# ungroup()

# sample 10 rows
slice_sample(species_rich, n = 10)
```

```
# A tibble: 40 x 3
# Groups:   year [5]
   year transect_name spp_rich
  <dbl> <chr>         <int>
1  2021 Pool 6             13
2  2021 Pool 1             13
3  2021 Pool 5             12
4  2021 Pool 2             14
5  2021 Pool 7             17
6  2021 Pool 4             15
7  2021 Pool 3             19
```

```

8 2021 Pool 8 19
9 2022 Pool 6 20
10 2022 Pool 3 18
# i 30 more rows

```

```

species_rich_final <- species_rich |>
  group_by(year) |>
  mutate(mean_pool = mean(spp_rich)) |>
  ungroup()

species_rich_final$year <- as.factor(species_rich_final$year)

```

Vegetation visualization

```

colors <- scale_color_manual(
  values = c(
    "Pool 1" = "#285943",
    "Pool 2" = "#D36582",
    "Pool 3" = "#82204A",
    "Pool 4" = "#74546A",
    "Pool 5" = "#C490D1",
    "Pool 6" = "#FFB7C3",
    "Pool 7" = "#5FB0B7",
    "Pool 8" = "#6C91BF"
  )
)

species_rich_final$year <- as.numeric(species_rich_final$year)

```

```

p <- ggplot(data = species_rich_final,
  aes(x = year,
    y = spp_rich,
    colour = transect_name,
    fill = transect_name
  )) +

# Geoms
geom_line(size = 1.7,
  alpha = 0.9) +
gghighlight(
  use_direct_label = FALSE,

```



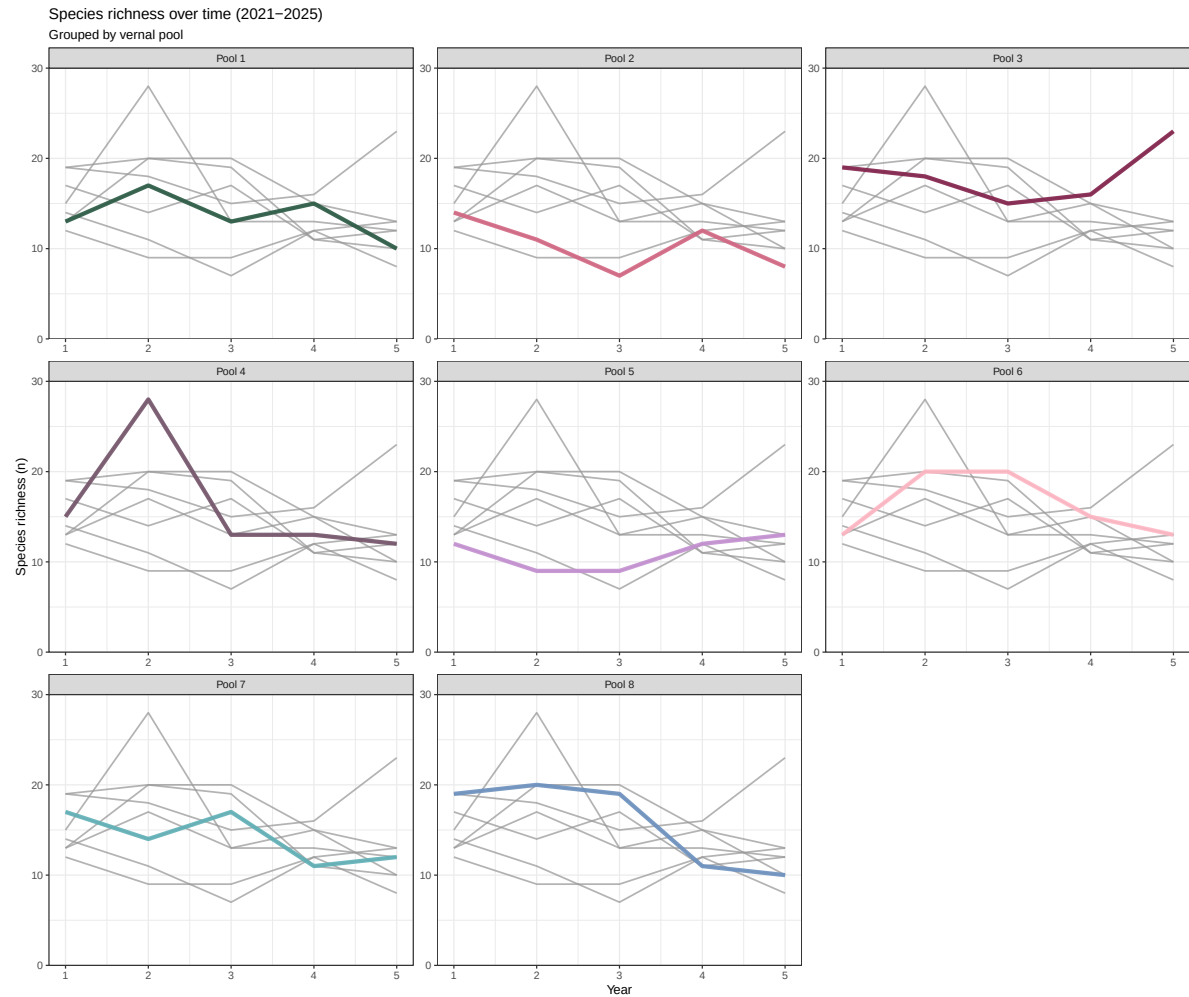
```

unhighlighted_params = list(
  linewidth = 0.7,
  alpha = 0.75,
  color = "gray60"
)
) +
# Scales
colors +
scale_y_continuous(
  expand = c(0, 0),
  limits = c(0, 30)
) +
# Labs
labs(
  title = "Species richness over time (2021-2025)",
  subtitle = "Grouped by vernal pool",
  caption = "Data derived from ENVS193DD course, UCSB",
  x = "Year",
  y = "Species richness (n)"
) +
# Facets
facet_wrap(~transect_name,
  scales = "free", ncol = 3
) +
# Theme
theme(
  plot.title = element_text(
    face = "bold",
    lineheight = 1.1,
    margin = margin(t = 5, b = 5)
  ),
  plot.subtitle = element_text(
    lineheight = 1.2,
    margin = margin(t = 5, b = 20)
  ),
  plot.caption = element_markdown(
    hjust = 0,
    margin = margin(t = 15, b = 5, l = 0),
    lineheight = 1.3
  ),
  strip.text = element_text(
    size = 12,
    face = "bold",

```

```
    color = "gray20",
    margin = margin(t = 5, b = 5)
  ),
  strip.background = element_rect(
    fill = "gray95",
    color = "white",
    linewidth = 0.5
  ),
  panel.spacing = unit(1.5, "lines"),
) +
theme_bw() +
theme(legend.position = "none")
```

p



Invertebrates

Invertebrate code

```
# new object from aq_ins_clean
log_aq_ins <- aq_ins_clean |>
  # mutate to get taxon group -species
  mutate(taxon = case_when(
    str_detect(taxon, "annelida") ~ "Annelida",
    str_detect(taxon, "cladocera") ~ "Cladocera",
    str_detect(taxon, "copepod") ~ "Copepod",
```

```

    str_detect(taxon, "diptera") ~ "Diptera",
    str_detect(taxon, "hemiptera") ~ "Hemiptera",
    str_detect(taxon, "nematode") ~ "Nematode",
    str_detect(taxon, "ostracod") ~ "Ostracod"
  )) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  subset(!is.na(med_do)) |>
  arrange(desc(log_ave_lit))

# sample of ten rows
slice_sample(log_aq_ins, n = 10)

```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	Cope~	5.87	9.72	74.2	11.4
2	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	Dipt~	0.267	9.29	NA	9.83
3	2023-11-12_NMC	2023-11-12 00:00:00	NMC	Clad~	0	NA	47	14.0
4	2022-11-12_NEC	2022-11-12 00:00:00	NEC	Dipt~	0	7.33	7.01	2.4
5	2023-03-03_NDC	2023-03-03 00:00:00	NDC	Nema~	0	7.45	0.688	5.64
6	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	Anne~	4.4	8.86	45.2	10.2
7	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	Anne~	0	7.71	1.98	9.59
8	2022-11-12_NEC	2022-11-12 00:00:00	NEC	Anne~	0	7.33	7.01	2.4
9	2022-05-11_NDC	2022-05-11 00:00:00	NDC	Ostr~	0	9.11	0.398	9.4
10	2021-11-16_NPB	2021-11-16 00:00:00	NPB	Ostr~	72.1	7.3	2.38	12.1

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>

```
view(log_aq_ins)
```

Invertebrate visualization

```

# creating colors object for ggplot
colors2 <- scale_color_manual(
  values = c(
    "Devereux Creek" = "#07beb8",

```

```

    "North of Venoco Bridge" = "#3dccc7",
    "South of Phelps Bridge" = "#68d8d6",
    "Phelps Road" = "#9ceaef",
    "South of Venoco Bridge" = "#c4fff9",
    "North of Phelps Bridge" = "#b2f7ef",
    "Main Channel" = "#bee9e8"
  ))

```

```

p2 <- ggplot(data = log_aq_ins,
  aes(x = taxon,
    y = log_ave_lit,
    colour = site_full,
    fill = site_full
  )) +

  # geometries
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  # adding median points
  stat_summary(geom = "point",
    fun = "median",
    size = 3,
    color = "darkblue") +
  gghighlight(
    use_direct_label = FALSE,
    unhighlighted_params = list(
      linewidth = 0.7,
      alpha = 0.75,
      color = "gray60"
    )
  ) +
  # Scales
  colors2 +
  scale_y_continuous(
    expand = c(0, 0),
    limits = c(-0.3, 7)
  ) +
  # Labs
  labs(
    title = "Median Aquatic invertebrate abundance/L\nat NCOS sites
    ↵ (2020-2024)",
    subtitle = "Median abundance as blue points",
    caption = "Data derived from ENV5193DD course, UCSB",
    x = "Taxon",

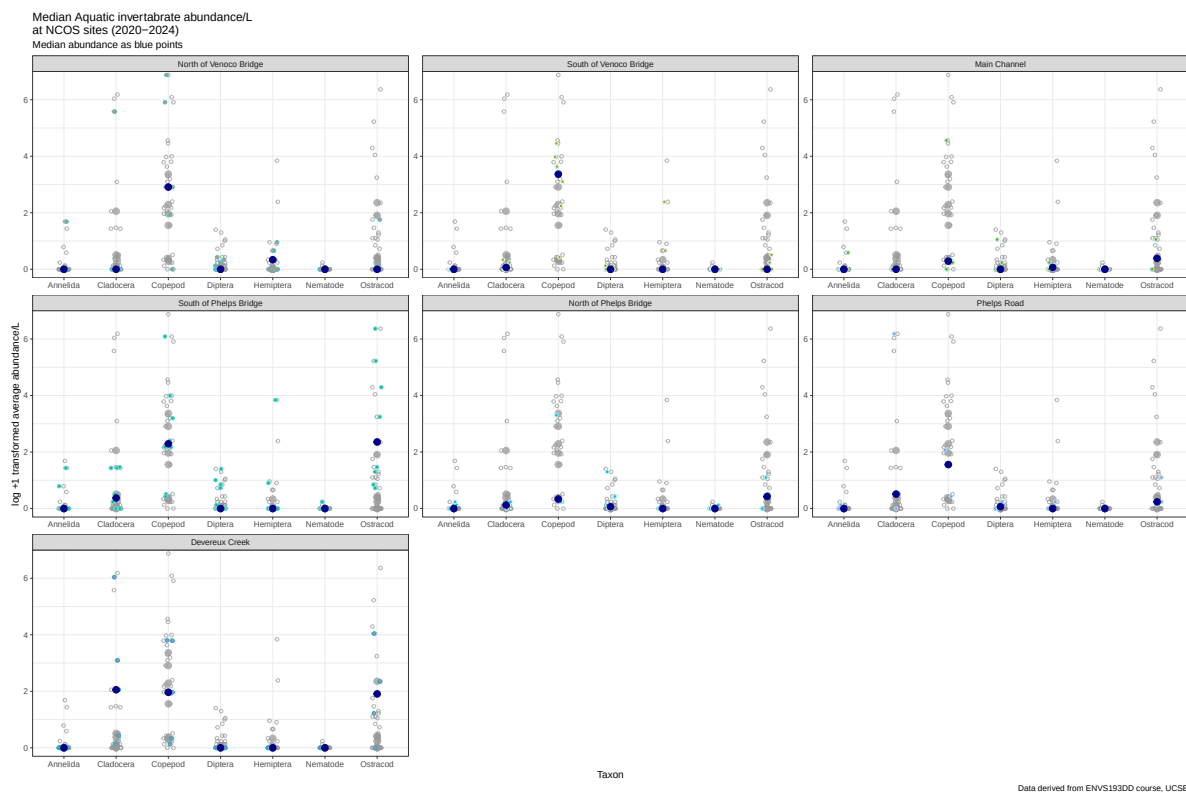
```

```

    y = "log +1 transformed average abundance/L"
  ) +
  # Facets
  facet_wrap(~site_full,
    scales = "free", ncol = 3
  ) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 10)) +
  # Theme
  theme(
    plot.title = element_text(
      face = "bold",
      lineheight = 1.1,
      margin = margin(t = 5, b = 5)
    ),
    plot.subtitle = element_text(
      lineheight = 1.2,
      margin = margin(t = 5, b = 20)
    ),
    plot.caption = element_markdown(
      hjust = 0,
      margin = margin(t = 15, b = 5, l = 0),
      lineheight = 1.3
    ),
    strip.text = element_text(
      size = 12,
      face = "bold",
      color = "gray20",
      margin = margin(t = 5, b = 5)
    ),
    strip.background = element_rect(
      fill = "gray95",
      color = "white",
      linewidth = 0.5
    ),
    panel.spacing = unit(1.5, "lines"),
  ) +
  theme_bw() +
  theme(legend.position = "none")

```

p2



Putting it all together

p / p2

4. In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective